MAHESH SINGH

MachineLearningEngineer | MLOps | GenAI/LLM Systems
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PROFESSIONAL SUMMARY

Final-year B.E. AI & Data Science student (CGPA: 8.3/10) with hands-on experience building drift-monitored ML pipelines, a hybrid log classification system, a market-crash detection model, and an open-source MCP tool for LLM context optimization. Interned at Celebal Technologies, improving model accuracy by 12% and cutting inference latency by 25%. Strong focus on MLOps, reproducible experimentation, and scalable deployment.

TECHNICAL SKILLS:

Programming: C,Python, SQL
Machine Learning: Scikit-learn, Pandas, Numpy, Tensorflow, Feature Engineering, Hyperparameter Tuning
Deep Learning & NLP: TensorFlow, Sentence Transformers, Sentence Embeddings
MLOps & Deployment: MLflow, Experiment Tracking, Data Drift Detection, KS Test, PSI, CI/CD, Docker, FastAPI
Data Visualization/Tools: Firebase, Matplotlib, Seaborn, Altair, Power BI, Excels
GENAI/LLM Systems: Gemini API, Groq API, LangChain, Prompt Engineering, RAG Architecture, Model Context Protocol(MCP), VectorLess Rag

EXPERIENCE

ML Intern — Infosys Springboard | Flash Crash Watchdog **Jun 2026 – Aug, 2026**

- Built Flash Crash Watchdog, a machine learning-based early-warning system to detect flash crashes and abnormal failure patterns in market data in near real time.
- Engineered market-behavior features and trained/compared SVM and Logistic Regression classifiers to flag high-risk crash windows, selecting the best -Performing model for deployment.

Data Science Intern – Celebal Technologies **Jun 2025 – Aug 2025**

- Designed and optimized supervised ML models on 50K+ structured records, improving classification accuracy by 12% via advanced feature engineering and hyperparameter tuning.
- Implemented MLflow-based experiment tracking and cross-validation framework, reducing model iteration cycle time by 30%.
- Reduced inference latency by 25% through batch-processing optimization and model simplification strategies.

IIT Guwahati – Summer Analytics 2025 (6 Weeks)

- Built end-to-end ML pipeline with preprocessing, feature scaling, and classification, improving model performance.

Tech Lead — Kaizen (M.B.M University Club) **Sept 2025 - Present**

- Mentor 35+ students on applied ML projects and system design principles.

FLAGSHIP PROJECTS

Self-Healing ML Pipeline (Drift Detection & Auto-Retraining) [Live Link]

- Designed automated pipeline with KS-Test & PSI drift detection ensuring sustained 92%+ accuracy. Triggered conditional retraining workflows reducing manual monitoring effort by 70%.

ShadowShield MCP — Open-Source npm Package [Live Link]

- Built a local-first MCP optimization layer for Claude Desktop and Cursor, reducing redundant LLM context through tool-call deduplication, response compression, and prompt/code optimization.
- Engineered SQLite-backed deterministic caching and rule-based output compression, with all-MiniLM-L6-v2 cosine-similarity

Hybrid Log Classification System (94.5% Accuracy) [Live Link]

- Built hybrid classification pipeline processing 100K+ log entries using rule-based Regex filtering and semantic
- Embeddings (Sentence Transformers) with Logistic Regression classifier.
- Achieved 94.5% accuracy and reduced manual log triage workload by 60%.

EDUCATION

B.E. Artificial Intelligence & Data Science – M.B.M University (2024–2027) **CGPA: 8.30/10**
Coursework: Probability, Statistical Inference, Machine Learning, Data Structures & Algorithms
Diploma in Computer Science – Govt. Polytechnic College (2021–2024) **CGPA: 9.71/10**

ACHIEVEMENTS

Abhivyakti Hackathon - Top 5 [Certificate]

Developed a Sentiment Analysis system to detect depression and emotional distress among students in coaching-heavy cities, using NLP-based emotion classification. Issues like depression and suicidal tendencies, particularly in cities like Kota, Rajasthan.

DevSummit25 Hackathon - Top 15 [Certificate]

Developed a cross-platform search engine using Gemini and DeepSeek APIs at a national-level hackathon.

Code For Bharat

Participated in Code for Bharat – Season 2 Hackathon (2025), developing and presenting a solution in a competitive national-level innovation environment.